

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, a circle has center C with coordinates (h, k) . Points A and B lie on the circle. Point A has coordinates $(h + 1, k + \sqrt{102})$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

Answer

- A. $\sqrt{206}$
- B. $2\sqrt{102}$
- C. $103\sqrt{2}$
- D. $103\sqrt{3}$

Correct Answer: A

Rationale

Choice A is correct. It's given that points A and B lie on the circle with center C . Therefore, $\overline{AC}$ and $\overline{BC}$ are both radii of the circle. Since all radii of a circle are congruent, $\overline{AC}$ is congruent to $\overline{BC}$. The length of $\overline{AC}$, or the distance from point A to point C , can be found using the distance formula, which gives the distance between two points, (x_1, y_1) and (x_2, y_2) , as $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$. Substituting the given coordinates of point A , $(h + 1, k + \sqrt{102})$, for (x_1, y_1) and the given coordinates of point C , (h, k) , for (x_2, y_2) in the distance formula yields $\sqrt{(h + 1 - h)^2 + (k + \sqrt{102} - k)^2}$, or $\sqrt{1^2 + (\sqrt{102})^2}$, which is equivalent to $\sqrt{1 + 102}$, or $\sqrt{103}$. Therefore, the length of $\overline{AC}$ is $\sqrt{103}$ and the length of $\overline{BC}$ is $\sqrt{103}$. It's given that angle ACB is a right angle. Therefore, triangle ACB is a right triangle with legs $\overline{AC}$ and $\overline{BC}$ and hypotenuse $\overline{AB}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting $\sqrt{103}$ for a and b in this equation yields $(\sqrt{103})^2 + (\sqrt{103})^2 = c^2$, or $103 + 103 = c^2$, which is equivalent to $206 = c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{206} = c$. Therefore, the length of $\overline{AB}$ is $\sqrt{206}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the length of $\overline{AB}$ if the length of $\overline{AC}$ were 103 , not $\sqrt{103}$.

Choice D is incorrect and may result from conceptual or calculation errors.